Investments & Loans Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

1 Electricity prices have risen by 3.4% p.a. over the past few years. If the average electricity bill in 2012 was $425, what was the average electricity bill in 2020? 2

2012 → 2020 is 8 years ∴ $n = 8$

$r = 3.4\% = 0.034$

$$FV = PV(1+r)^n$$
$$= 425(1+0.034)^8$$
$$= 425 \times 1.034^8 = \$555.33$$

2 Sophia purchased 1500 shares for $3.25 and later sold them for $4.90 per share. Sophia paid brokerage fees on the purchase and sale of the shares as shown in the table. Given that she did not receive any dividend, calculate the profit that Sophia earned from the shares. 3

Brokerage fees	
Purchase of shares	2% of the purchase price
Sale of shares	3% of the sale price

Purchase price = 1500 × $3.25
= $4875

Total purchase cost = purchase price + brokerage
= $4875 + 0.02 × 4875
= $4875 + $97.50
= $4972.50

Sale price = 1500 × $4.90 = $7350

Total income from selling = sale price − brokerage
= $7350 − 0.03 × 7350
= $7350 − $220.50
= $7129.50

Profit = $7129.50 − $4972.50
= $2157

3 Grace invested $32 000 at a simple interest rate of 7% pa. Calculate the future value of Grace's investment after 5 years. 1

$$32000 \times 0.07 \times 5 + 32000 = \$43200$$

4 Last month Lenny bought 220 shares at a market price of $1.35 per share. The market price is now $1.37 per share. Lenny received a dividend of 8.5 cents per share. What is the dividend yield correct to 1 decimal place? 1

$$\frac{0.085}{1.37} \times 100 = 6.2\%$$

5 A piece of artwork is currently valued at \$14 500. It is estimated that the artwork will appreciate at an annual rate of 2.5% over the next 4 years. Which of these calculations would give the estimated value of the artwork in 4 years' time? 1

A. $14\,500 \div 1.025 \times 3$

B. $14\,500 \times 1.0.25 \times 3$

C. $14\,500 \div 1.025^4$

D. $14\,500 \times 1.025^4$

$$14\,500 \times 1.025^4$$

6 Fran's credit card has a no interest-free period. Interest is charged at 0.039% compounding per day. Interest is charged on the date of purchase but not the date the account is paid. In June, Fran made only one purchase on the 7th of June, a pair of shoes for \$125. Fran paid the account on the 1st of July. Calculate the total interest charged for June. 3

7th June to 30th June is 24 days.

$$FV = PV(1+r)^n$$
$$= 125(1+0.00039)^{24} = 126.17526\ldots$$
$$\approx \$126.18$$
$$I = FV - PV = 126.18 - 125 = \$1.18$$

Interest charged is \$1.18

7 In 2019, Jasmine bought a home for \$875 000. Over the next 3 years, the house appreciated by 6.5% p.a. Determine how much the value of the house had increased by over the 3 years, correct to the nearest \$1000. 3

$$FV = 875000(1+0.065)^3$$
$$= \$1056955.922\ldots$$
$$\text{Increase amount} = 1056955.922 - 875000$$
$$= \$181\,955.9219\ldots = \$182\,000$$
(to the nearest \$1000)

8 The following table shows the monthly repayments for each \$1000 borrowed. Davy is planning to borrow \$1 220 000 for a house and to repay the loan over a period of 30 years. The bank has offered him an interest rate of 4% per annum. How much interest will he end up paying over the term of the loan? 2

	Interest Rate per Year		
Time Period (Years)	4%	5%	6%
20	\$6.06	\$6.60	\$7.16
25	\$5.28	\$5.85	\$6.44
30	\$4.77	\$5.37	\$6.00

$$\text{Total Loan} = 1220 \times 4.77 \times 30 \times 12$$
$$= \$2094984$$
$$\text{Interest} = 2094984 - 1220000$$
$$= \$874\,984$$

9 Shaw received this credit card statement from CBA for the month of May. Interest was charged at rate of 18.98% per annum, compounding daily. There is no interest-free period. The period for which interest was charged included the date of purchase and the date of payment. What amount was paid when the account was paid in full on 31 May? 3

Purchase Date	Transaction Details	Purchase Amount
08 May	Booktopia Pty Ltd Lidcombe	\$33.90
21 May	NRMA Insurance	\$1906.84

$$\text{Amount paid} = 33.90\left(1+\frac{0.1898}{365}\right)^{24}$$
$$+1906.84\left(1+\frac{0.1898}{365}\right)^{11}$$
$$= \$1952.10$$

10 The table below shows the repayment per \$1000 on a monthly reducible loan.

Term in Years	7%	7.25%	7.5%	7.75%	8%
5	19.8012	19.9194	20.0379	20.1570	20.2765
10	11.6108	11.7401	11.8702	12.0011	12.1328
15	8.9883	9.1286	9.2701	9.4128	9.5566
20	7.7530	7.9036	8.0559	8.2095	8.3644
25	7.0678	7.2281	7.3899	7.5533	7.7182
30	6.6530	6.8218	6.9921	7.1641	7.3377

a. Use the table above to find the monthly repayment per \$1000 on a loan borrowed at 7% for 20 years. 1

7.7530

b. Calculate the monthly repayment on a loan of \$70000 at 7% over 20 years. 2

$70 \times 7.7530 = 542.71$

c. Hence, calculate the total amount repaid over the term of the loan in part (b)? 1

$542.71 \times 20 \times 12 = 130250.4$

d. Determine how much interest was paid after the loan in part (b) was fully paid? 1

$130250.4 - 70000 = \$60250.4$

11 Jimmy needs \$1000 in 5 years' time. He is going to invest some money today in an account earning 3% per annum compounded annually. He will make no further deposits or withdrawals. How much money he needs to invest today so that he has \$1000 at the end of 5 years at 3% p.a compounded annually. 2

$A = P(1+r)^n$

$P = \dfrac{1000}{(1+3\%)^5} = 862.608$

12 Last month, Jessica purchased 550 shares at a market price of \$4.00 per share. The market price is now \$4.60 per share. Jessica receives a dividend of \$0.40 per share. What is the dividend yield, to the nearest percent? 1

$\frac{0.40}{4.60} \times 100$ 9%

13 Scarlett uses a credit card that has an interest rate of r % per annum, simple interest, daily. The card does not have an interest-free period. Interest is calculated to include the day of transaction and the day on which the account balance is paid. During December, Scarlett makes the following transactions. 3

Credit card statement

Date	*Transaction*	*Amount*
12 December	clothing store	\$84.00
19 December	shoe store	\$150.00
21 December	supermarket	\$170.00

If Scarlett pays her account balance of \$407.05 on 5 January in full, what is the annual interest rate, to 2 d.p.?

$407.05 - 404 = 3.05$

$(84 \times 25 \times r) + (150 \times 18 \times r) + (170 \times 16 \times r) = 3.05$

$2100r + 2700r + 2720r = 3.05$

$7520r = 3.05$

$r = 0.0004\ldots \times 365 \times 100$

rate p.a. = 14.80 %

Investments & Loans Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

1 After 5 years, the value of a piece of farm machinery has reduced to \$25 000. The original value of the machinery when purchased new was \$80 000.

a. If the straight-line method was used to depreciate the machinery, what was the annual loss in value as a percentage of the purchase price? 2

$25000 = 80\,000 - 5D$

$D = 11\,000$

as a % $\frac{11000}{80000} \times 100 = 13.75\%$

b. Show that if the declining balance method was used to depreciate the machinery, the annual loss in value as a percentage of the purchase price was 20.76%. 2

$25000 = 80\,000(1-r)^5$

$(1-r)^5 = 0.3125$

$1-r = \sqrt[5]{0.3125}$

$1-r = 0.792\ldots$

$r = 0.2075\ldots$

$\approx 20.76\%$ p.a

2 Ollie just got their P's and is trying to save up to buy their first car. They have decided to contribute a certain amount from each monthly pay slip into a high interest savings account. Ollie has created an Excel spreadsheet to keep track of the monthly contributions. A screenshot of the spreadsheet is below:

	A	B	C	D	E	F	G
1	**End of month**	**P**	**I**	**P+I**	**P + I + R**		
2	**1**	500.00	1.00	501.00	1001.00		
3	**2**	1001.00	2.00	1003.00	1503.00		**interest rate p.a.**
4	**3**	1503.00	3.01	1506.01	2006.01		**2.40%**
5	**4**	2006.01	4.01	2010.02	2510.02		
6	**5**						
7	**6**						
8							

a. What amount has Ollie decided to contribute into the account each month? \$500 1

b. By continuing the calculations, what is the amount in cell E7? 2

$2510.02\left(1+\frac{0.024}{12}\right) + 500 = 3015.04$

$3015.04\left(1+\frac{0.024}{12}\right) + 500 = 3521.07$

c. What amount could Ollie have contributed initially (PV) into this account, without making any further contributions, to have the same financial outcome after 6 months? 2

$3521.07 = P\left(1+\frac{0.024}{12}\right)^6 \quad P = \frac{3521.07}{\left(1+\frac{0.024}{12}\right)^6} = \$3479.11$

3 Aaron wants to purchase an apartment. The bank manager has provided them with a table of monthly payments for a loan of $1000 for varying interest rates.

Interest rate per annum	*Term of loan in years*					
	5	*10*	*15*	*20*	*25*	*30*
6.0%	19.33	11.10	8.44	7.16	6.44	6.00
6.5%	19.57	11.35	8.71	7.46	6.75	6.32
7.0%	19.80	11.61	8.99	7.75	7.07	6.65
7.5%	20.04	11.87	9.27	8.06	7.39	6.99
8.0%	20.28	12.13	9.56	8.36	7.72	7.34

Aaron has savings of $95 000 to put towards the purchase price of his $620 000 apartment.

a. Aaron decides to purchase the apartment using $50 000 of his savings as a deposit and to borrow the balance over 20 years at a rate of 6.5% p.a. Find Aaron's monthly repayment. 2

620 000 − 50 000 = 570 000

570 × 7.46 = 4252.20

b. After reviewing their financial position with the bank manager, Aaron realises he can't afford to pay any more than $2200 per month. The bank will lend at 6.0% per annum over 30 years provided Aaron uses all of his savings as a deposit. The owner of the apartment is prepared to drop the sale price of the apartment by $10 000. Can Aaron afford this apartment? 2

610 000 − 95000 = 515 000

515 × 6 = 3090

∴ no, he can't afford i

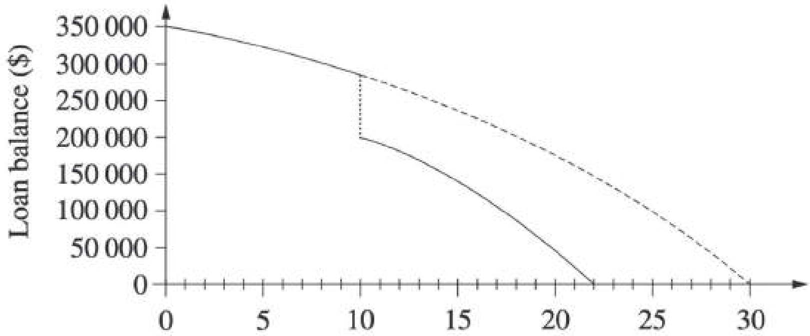
4 Nick bought a portfolio of 2000 MNRA shares with his retrenchment payout. The value of each share is currently $12.50, and Nick is paid an annual dividend of $0.75 per share. What is the dividend yield on the shares? 1

$\frac{0.75}{12.50} = 0.06$

5 A brand new computer which costs $2500 will depreciate at 30% per annum for the next 3 years using the declining balance method. What will it be worth at the end of three years? 2

Salvage $= V_0(1-r)^n = 2500(1-0.3)^3 = \$857.50$

6 Jamal borrowed \$350 000 to be repaid over 30 years, with monthly repayments of \$1880. However, after 10 years he made a lump sum payment of \$80 000. The monthly repayment remained unchanged. The graph shows the balances owing over the period of the loan. Over the period of the loan, how much less did Jamal pay by making the lump sum payment. 2



30 − 22 = 8 year, so 96 month

96 × 1880 = 180 480

180480 − 80000 = 100480

(lump sum)

7 Calculate the total cost of purchasing 3000 mining shares with a market price of \$5.85 if brokerage is 2·5%. 2

$3000(5\cdot85)(1\cdot025) = \$17\ 988\cdot75$

8 The table below gives the monthly repayments for a reducing balance loan of \$1000 for various terms and interest rates. What is the monthly repayment on a \$450 000 loan at 6%p.a. over 15 years? 1

Monthly repayments per $1000 borrowed						
Interest rate	Term (years)					
(% p.a.)	5	10	15	20	25	30
4.5	$18.64	$10.36	$7.65	$6.33	$5.56	$5.07
5.0	$18.87	$10.61	$7.91	$6.60	$5.85	$5.37
5.5	$19.10	$10.85	$8.17	$6.88	$6.14	$5.68
6.0	$19.33	$11.10	$8.44	$7.16	$6.44	$6.00
7.0	$19.80	$11.61	$8.99	$7.75	$7.07	$6.65
8.0	$20.28	$12.13	$9.56	$8.36	$7.72	$7.34

$450 \times 8\cdot44$

9 A car depreciates in value from \$78 000 to \$41 000 in 3 years. Use the declining-balance formula to calculate the annual percentage rate of depreciation. 2

$41\ 000 = 78\ 000(1-r)^3$

$(1-r)^3 = 0\cdot5256\ldots$

$1-r = 0\cdot80704\ldots$

$r \doteqdot 19\cdot3\%$ p.a.

10 Zoe borrowed $875 000 to purchase her first home. The bank lent her the money at 5·2%p.a. reducible interest and fortnightly repayments of $3444.65.

a. Complete the table below showing the progress of Zoe's loan for the first 6 fortnights. 4

Fortnights (n)	Principal (P)	Interest (I)	Amount owing ($P+I$)	Balance ($P+I-R$)
1	\$875 000	\$1750	\$876 750	\$873 305.35
2	\$873 305.35	\$1746.61	\$875 051.96	\$871 607.31
3	871 607·31	1743·21	873 350·52	869 905·87
4	869 905·87	1739·81	871 645·68	868 201·03
5	868 201·03	1736·40	869 937·43	866 492·78
6	866 492·78	1732·99	868 225·77	864 781·12

b. How much has she paid off the principal after six repayments? 1

875 000 − 864 781·12 = $10 218·88

c. How much interest did she pay in the first 12 weeks? 1

6 × 3444·65 − 10 218·88 = $10 449·02

11 Calculate the amount that must be invested at 3·2%p.a. interest compounding quarterly to have $10 000 at the end of 7 years. 1

$FV = PV(1+r)^n$

$10\,000 = PV\left(1 + \frac{0.032}{4}\right)^{7\times4}$

$PV = \frac{10\,000}{(1.008)^{28}}$

$= \$8000.28$